Topic 2.15 - Semi-log Plots

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________________ Class: ______________ Date: ______________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Interpret semi-log plots with a logarithmically scaled output axis.
- Use linearity in x versus $\log y$ to identify and construct exponential models.
- Explain how slope, intercept, base, shifts, and limitations affect linearization.

Section A: Foundation and representation

Question 1

Core skill | Calculator permitted

Linearize $y = 18(1.25)^x$ using common logarithms. Identify the slope and intercept.

Question 2

Core skill | Calculator permitted

A line on a plot of x versus $\ln y$ is $\ln y = 2.4 + 0.18x$. Write the exponential model.

Question 3

Core skill | Calculator permitted

A line on a base-10 semi-log plot is $\log y = -0.25x + 3$. Find the initial value and one-unit factor.

Question 4

Core skill | No calculator

Two points on a $\ln y$ versus x line are $(2, 3.1)$ and $(7, 4.6)$. Find the exponential model $y = ab^x$.

Question 5

Core skill | Calculator permitted

On a base-10 semi-log line, increasing x by 5 raises $\log y$ by 1.2. Find the factor for a one-unit increase and for a five-unit increase.

Question 6

Core skill | No calculator

A semi-log line has slope $\log 0.84$. What percent change occurs per one-unit increase in x?

Section B: Application, comparison, and error analysis

Question 7

Core skill | Calculator permitted

For the data below, compute $\log_{10} y$ and decide whether an exponential model is plausible.

x	0	1	2	3
y	20	40	80	160

Question 8

Shifted model | No calculator

A shifted model is $P(t) = 50 + 300(0.9)^t$. Describe the correct quantity to plot on a logarithmic scale to obtain a line.

Question 9

Error analysis | No calculator

A student reads equal vertical spacing between 10 and 100 and between 100 and 1000 as equal output differences. Correct the interpretation.

Question 10

Validation | No calculator

A data set looks linear on both an ordinary plot and a semi-log plot over only three points. What should be done before choosing a model?

Question 11

Base reasoning | No calculator

If $\log_2 y = 4 - 0.5x$, write y as an exponential and identify the factor per two-unit increase in x.

Question 12

Limitation | No calculator

A semi-log model fitted on $0 \leq x \leq 8$ is extrapolated to $x = 80$. State two reasons for caution.

Section C: AP-style multiple choice

MCQ 1

No calculator

If $\log_{10} y = 0.2x + 1$, the exponential factor per unit is:

A. 0.2

B. 10

C. $10^{0.2}$

D. 1.2

MCQ 2

No calculator

A straight decreasing line on a plot of x versus $\ln y$ suggests:

A. Linear growth in y

B. Exponential decay in y

C. Quadratic growth in y

D. A logarithmic model for y

MCQ 3

No calculator

For $y = L + ab^x$, which transformation is most likely to linearize the data?

A. Plot $y - L$ against x on ordinary axes

B. Plot $\log(y - L)$ against x

C. Plot $\log x$ against y

D. Plot $1/y$ against x

MCQ 4

No calculator

Equal vertical distances on a base-10 log axis represent equal:

A. Differences

B. Ratios

C. Slopes

D. Intercepts

Section D: Free-response reasoning

FRQ 1 - Construct from a semi-log line

Calculator permitted | 6 points

A fitted line on a plot of x versus $\log_{10} y$ is $\log y = 1.3 + 0.12x$. (a) Write the exponential model $y = ab^x$. (b) Interpret a and b. (c) Predict y at $x = 10$. (d) Find the input at which the model reaches $y = 500$.

FRQ 2 - Shifted exponential linearization

Calculator permitted | 6 points

A process is believed to follow $T(t) = L + Ab^t$. A known limiting value is $L = 18$. Measurements are $T(0) = 82$, $T(2) = 54$, and $T(4) = 38.25$. (a) Compute $T - L$ for each time. (b) Use the transformed values to test whether one exponential factor is plausible. (c) Construct a model. (d) Explain what would happen if $\log T$ were plotted instead of $\log(T - 18)$.